Astronaut Training



Does anyone know where this astronaut training center is?

Houston Texas!

Lesson 2:
Astronaut Selection & Training Facilities

Image credit: NASA

Recap from last week

- An astronaut is a person whose ***profession*** is to travel ***beyond the Earth's atmosphere***
- Space starts at Kármán line at 62 miles (or 100km)
- Astronauts reach space on rockets like the Saturn V
- There are many space programs:
  - Space shuttles (135 missions)
  - Soyuz program (157 crewd launches)

Can you name some famous astronauts?



Neil Armstrong
Buzz Aldrin
Michael Collins
Yuri Gagarin
Valentina Tereshkova

How the classes work

Red Boxes Show Activities

We will do a hands on activity!

Green Boxes are questions

Raise your hand if you want to answer!

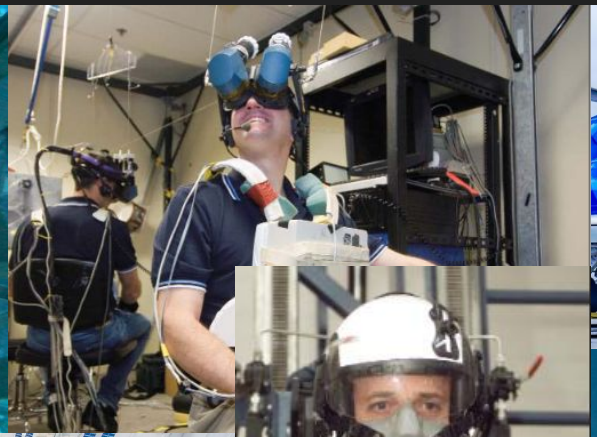
Blue Boxes are fun facts or answers

Here you learn something new!

- Raise your hand if you have a question.
- You are always free to ask your neighbour.
- Be respectful of others. Give them time and space to participate as well.
- Pay attention to the lecture. You will learn some really cool stuff!

Today's Topic: Astronaut selection













NASA Astronaut selection

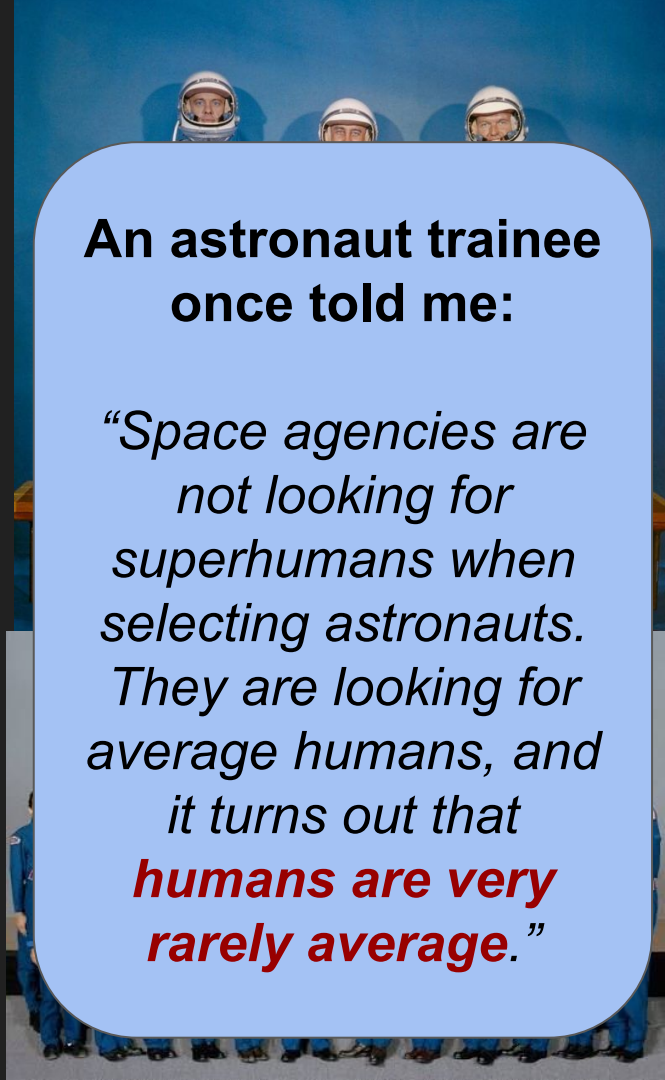
What do you think are requirements?





NASA Astronaut selection

- The requirements to become an astronaut are:
  - Must be U.S. citizen
  - Must have a bachelor's degree in engineering, biological science, physical science, or mathematics.
  - 3 years of experience (for example teaching K-12 or a PhD)
  - 20/20 vision (Glasses are allowed)
  - Good blood pressure (not more than 140/90)
  - Height between 5'2" and 6'3"



An astronaut trainee once told me:

"Space agencies are not looking for superhumans when selecting astronauts. They are looking for average humans, and it turns out that ***humans are very rarely average****."*

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Do you need to be a pilot to become an astronaut?

No you don't need to be a pilot anymore



NASA Astronaut selection

- NASA has calls for astronauts every few years. The last one was 2024.
- **Next application cycle is expected around 2030.**
- After a first selection, a week-long process of personal interviews, medical screening, and orientation take place to find the new astronauts

Image credit: NASA





How much do Astronauts earn?

Astronauts earn $152,258.00 per year

ESA Astronaut selection

- Screening: height, weight, eyes, ears
- Test Phase 1: cognitive, technical, and motor testing.
- Test Phase 2: individual exercises and practical tests.
- Test Phase 3: physical and mental abilities
- Interview Round 1: technical and behavioural competencies
- Interview Round 2: final interview with the ESA Director



ESA Astronaut selection

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- Interview Round 2: final interview with the ESA Director
- Last cycle, ESA specifically recruited an astronaut with a physical disability: John McFall



NASA Astronaut training

Group Discussion:

What do astronauts learn during training?

Skills:

- ?
- ?
- ?
- ?
- ?
- ?

Training environments:

In Houston TX:

- ?
- ?
- ?
- ?

Not in Houston:

- ?

NASA Astronaut training

Group Discussion:

What do astronauts learn during training?

Skills:

- Russian & English language
- Cultur & Law & Politics
- Medicine & Survival training
- Space systems and vehicles
- Orbital mechanics
- Scuba diving

Training environments:

In Houston TX:

- Neutral Buoyancy Lab (NBL)
- Space Vehicle Mock-up Facility (SVMF)
- The Precision Air-Bearing Floor (PABF)
- Active Response Gravity Offload System

Not in Houston:

- Microgravity flights

Russian

Space Law

Medicin

Russian | Space Law | Medicin



Russian



Hello (informal)
pree-vet

Hell (formal)
zdras-tvuy-te

Space Law

Can the military build a fortress in space?
No, space can only be used for peaceful purposes.

Can you own a piece of land on the Moon?
No, space belongs to all, no one can claim ownership.

Who is responsible if SpaceX commits a crime in space?
The US government

Medicin

Russian



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Medicin

If you cut your finger, what should you do?
Disinfect the wound and put on a band aid.

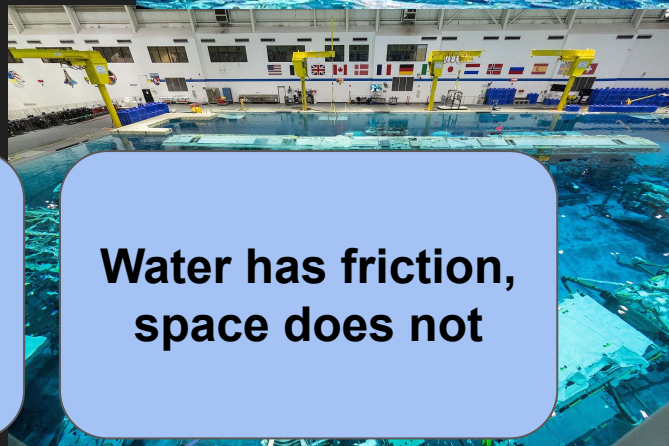
What should you do if you start to panic?
Take slow and deep breaths, or meditate.

Which is an astronaut most likely to hurt?
Shoulder, back, hips, knees
Shoulders!

Neutral Buoyancy Laboratory

- Water is used to simulate the weightlessness of space and the lack of oxygen
- The training pool has a complete space station underwater to train for spacewalks (bottom image)
- Here they also test the return capsules because they will land in water (top image).





What are the two most important difference between weightlessness in water and in outer space?

Water has high pressure, space has low pressure

Water has friction, space does not



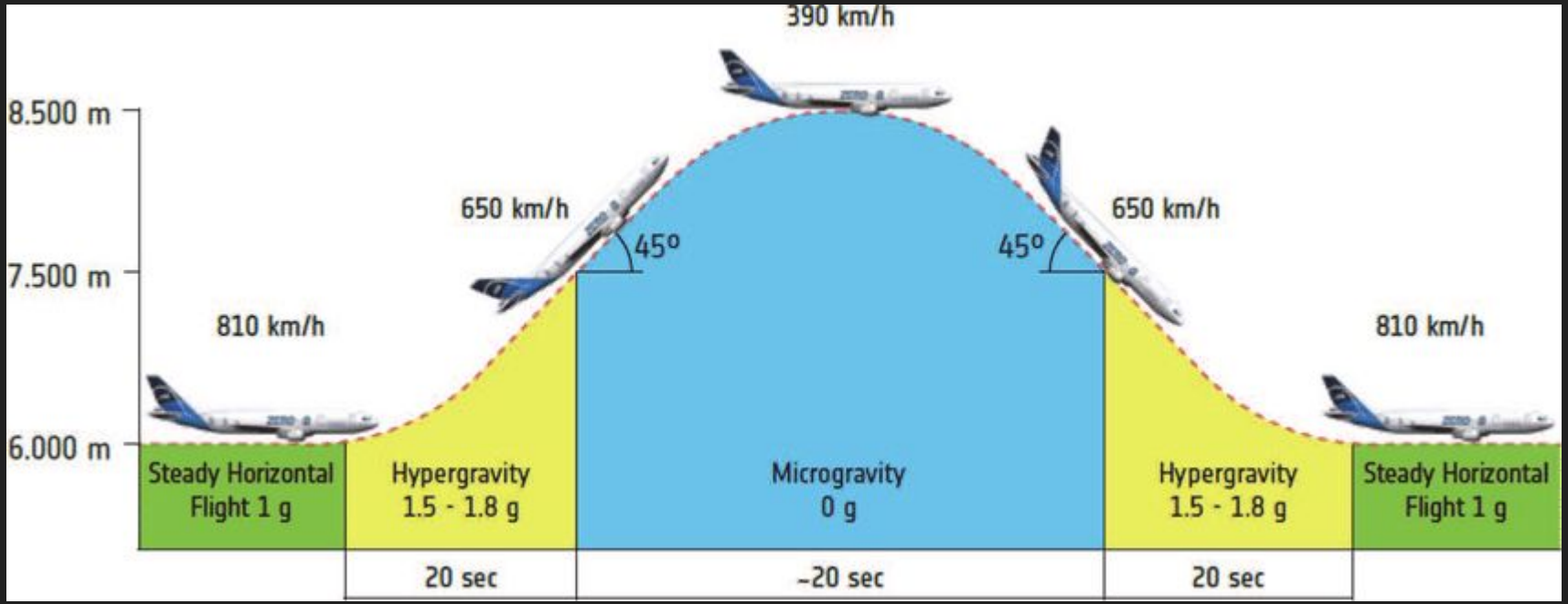
Parabolic Flights

- **Parabolic Flights mimic weightlessness**
- Astronauts learn how to handle microgravity
- They learn how to conduct experiments and use space systems
- This is difficult because humans are used to gravity (remember the no friction floor!)





Parabolic Flights



Step 1: The plane accelerates upwards very quickly

Step 2: The plane shuts of the engines and is now in free fall

Step 3: The engines are turned back on and the plane is catched


esa
ZERO GRAVITY SCIENCE

Parabolic Flights

- Parabolic Flights are also available commercially
- A single seat costs $9000
- Renting the whole plane costs $260000
- If you get motion sick, you should stay home
- Since you can rent them, we get amazing things like these ….



Image credit: Zero-G


vevo

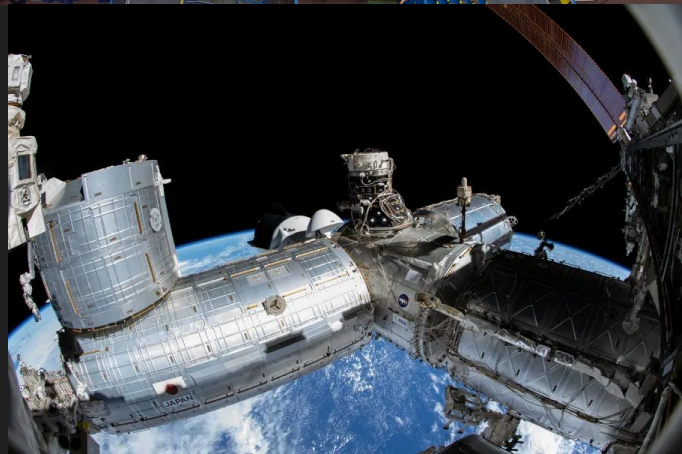
Space Vehicle Mock-up Facility (SVMF)

- At SVMF the astronauts learn how to operate space systems and vehicles
- They have life sized mock ups of many different space crafts and modules
- Human Exploration Research Analog (HERA) for isolation during space travel



Do you recognize what the module on the top right is for?

The International Space Station!



Space Vehicle Mock-up Facility (SVMF)

- Crew Health and Performance Exploration Analog (CHAPEA)
- 4 crew members will spend 378 days enclosed in CHAPEA
- This work is in preparation for prolonged stays on Mars
- This is the floor plan:





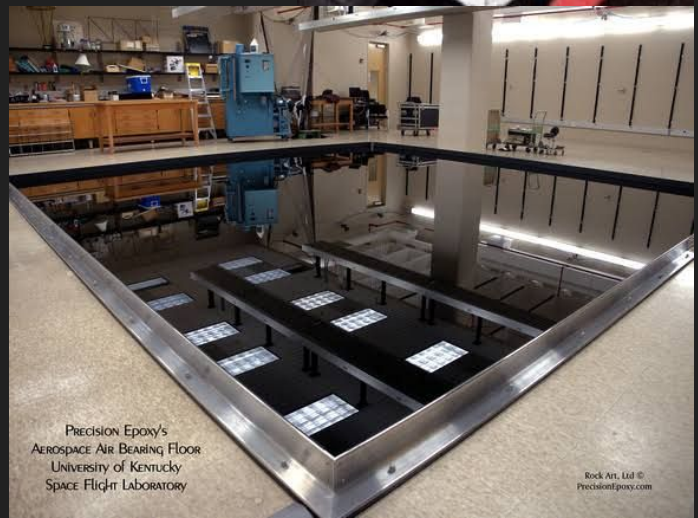


Raise your hand if you would like to live here for a year

The Precision Air-Bearing Floor (PABF)

- PABF is a no-friction floor
- It works on the same principle as air hockey tables but instead of the table, the puck is blowing air
- Here astronauts learn how to move large objects when there is no friction
- This is very difficult since humans are used that things stop when we place them down
- On this floor, they just continue moving





Active Response Gravity Offload System

- ARGOS pulls you up with a constant force
- This has the same effect as lower gravity
- Astronauts can learn how to walk on the Moon
- It works the same as an automatic auto belay for climbing

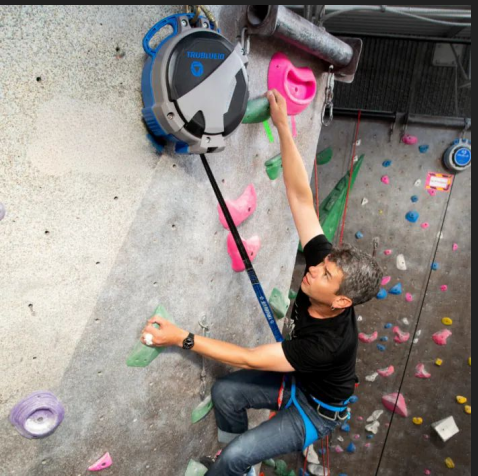
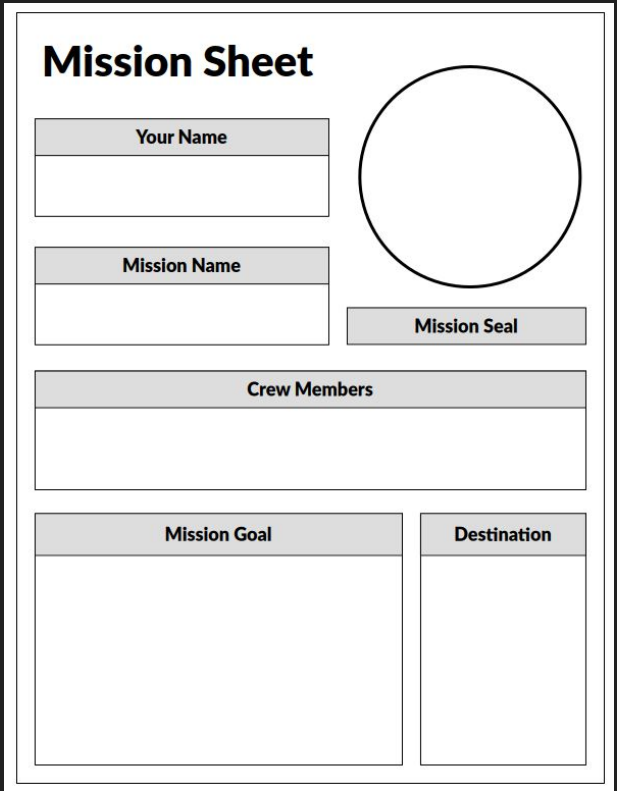


Image credit: TruBlue



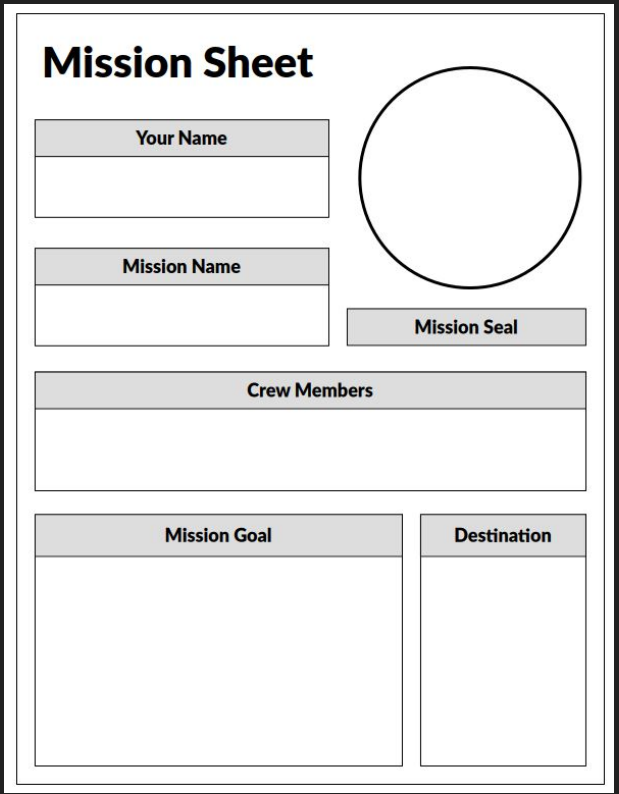


Today's Activity: Plan Your Space Mission



- Split into groups of 4
- Fill in your name
- Find a cool mission name
- Write down your crew members
- What do you want to achieve?
- Where do you want to travel to in space?
- **Don't forget to draw a cool mission seal!**

Homework



Mission Sheet

Your Name

Mission Name

Mission Seal

Crew Members

Mission Goal

Destination

- Finish your mission sheet
- In particular the mission seal!

Next Week: Living in Space

